



Superiorly low half-wave voltage electro-optic polymer modulator for visible photonics

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Abstract: Chip-scale optical devices operated at wavelengths shorter than communication wavelengths, such as LiDAR for autonomous driving, bio-sensing, and quantum computation, have been developed in the field of photonics. In data processing involving optical devices, modulators are indispensable for the conversion of electronic signals into optical signals. However, existing modulators have a high half-wave voltage-length product ($V_{\pi}L$) which is not sufficient at wavelengths below 1000 nm. Herein, we developed a significantly efficient optical modulator which has low $V_{\pi}L$ of 0.52 V·cm at $\lambda = 640$ nm using an electro-optic (EO) polymer, with a high glass transition temperature ($T_g = 164$ °C) and low optical absorption loss (2.6 dB/cm) at $\lambda = 640$ nm. This modulator is not only more efficient than any EO-polymer modulator reported thus far, but can also enable ultra-high-speed data communication and light manipulation for optical platforms operating in the ranges of visible and below 1000 nm infrared.

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1. Introduction

The growth of global internet traffic has led to the demand for optical networks with high-performing electro-optic (EO) modulators for the conversion of electrical signals to light waves. Modulators are one of the most important components in the optical signal processing of integrated circuits, and are the most effective at high speeds and low driving voltages. While modulators have traditionally operated at communication wavelengths above the 1000-nm near-infrared range [1–5], photonic platforms operating at wavelengths below 1000 nm have recently begun to attract attention in fields such as optogenetics [6–10], bio-sensing [11–16] and quantum optics [17–20]. Moreover, an optical phased array (OPA) has been developed in the visible range of the photonic platform [21] and is particularly used in the light detection and ranging (LiDAR) of high-speed operations, which is required for autonomous driving and 3D cameras. LiDAR primarily operates at wavelengths below 1000 nm, where an inexpensive Si photodetector can be used. In addition, the deterioration of signals at wavelengths below 1000 nm (such as $\lambda = 905$ nm) due to rain, fog, and snow is lesser than that at near-infrared wavelengths (such as $\lambda = 1550$ nm) — the water extinction coefficient at $\lambda = 905$ nm is 145 times lower than that at $\lambda = 1550$ nm [22]. For this reason, OPAs with an operating wavelength of 905 nm [23] are preferable. For these optical devices, it is necessary to use a high-efficiency modulator operated at wavelengths below 1000 nm.

Si modulators, reported and demonstrated largely in the telecommunication optical wavelength bands [24–26], are highly compatible with Si photonic platforms. However, because the Si transparency window is limited to the telecommunication wavelength region above 1000 nm, other materials need be explored for wavelengths below 1000 nm. In the visible-wavelength region, lithium niobate (LN) and Si nitride (SiN) have been investigated as waveguide candidates for modulators. LN modulators have been operated in the visible to near-infrared region [27–29]; LN exhibits EO effects based on the Pockels effect and has properties suitable for practical applications, such as a wide transparency window and good temperature stability [30]. However,

LN modulators also exhibit undesirable properties such as a small refractive index change of 10^{-5} , data rates up to a modest modulation speed of 40 GHz, and difficulties in fine processing. Similarly, SiN has a wide transparency window from blue to near-infrared wavelengths; it also displays relatively small changes in its refractive index, which is inefficient because a long waveguide is required to obtain a sufficient change in the phase of light. However, its modulators operate across a wide range of wavelengths [31,32], based on EO or thermo-optic effects.

In this study, we focused on a modulator using an organic EO polymer with excellent optical properties. These polymers experience the Pockels effect owing to which they exhibit excellent EO properties, such as a significantly high EO coefficient of ~ 100 pm/V and a high-frequency response of over 100 GHz. For these reasons, the EO-polymer modulator is a promising candidate for high-speed optical communication. High-speed modulators [33] and OPAs [34] using EO polymers have been demonstrated at wavelengths of 1550 [35–38] and 1310 nm [36,37]. Unlike inorganic materials such as LN and Si, these EO polymers offer scientists the option to customize the transparency window across the visible to near-infrared region by designing the molecular structure. In contrast, there are few reports on the demonstration of EO-polymer modulators operated at wavelengths below 1000 nm; this is because efficient EO polymers that have a sufficient transparency window at wavelengths below 1000 nm have not been developed.

Thus far, efficient EO polymers that have either the phenyl-vinylene-thiophene-vinylene (FTC) [39] chromophore with thienyl-di-vinylene bridges or the phenyltraene-bridged CLD chromophore [40] have been extensively investigated for telecom applications. Because push-pull molecules with a long chain of π -conjugated groups (such as the FTC and CLD chromophores) have a strong resonance peak at wavelengths below 1000 nm, EO polymers with these chromophores can be used only at wavelengths above — but not at those below — 1000 nm. For applications operating at wavelengths below 1000 nm, EO polymers with the well-known Disperse Red 1 (DR1) chromophore — a relatively short push-pull molecule with a resonant peak at 500 nm [41] — may be used; however, they have a relatively broad absorption band and suffer large absorption losses at wavelengths below 1000 nm [42], which is a serious problem for applications with long propagation lengths, such as modulator devices. In addition, there is a report of modulation at $\lambda = 640$ nm when the DMABI-Ph6 chromophore is used, but the r_{33} value of its EO coefficient is 0.2 pm/V [43], which is significantly small.

In this study, we identified unique EO polymers that can be used at visible wavelengths. A novel side-chain polymer, with an EO chromophore in which a tricyanofuran (TCF) acceptor and an aminobenzene donor are directly connected by a single bond, was also synthesized in this study. Compared with the DR1 chromophore, the developed EO chromophore exhibits a similar absorption peak wavelength, equivalent hyperpolarizability [44], and a larger static dipole moment, which is important for efficient poling. The most interesting and spectacular optical characteristic of our EO polymer is its narrow absorption band, which is accompanied by a significantly sharp decrease in the longer wavelength region. Thus, our EO polymer exhibits a low absorption loss at an operating wavelength of 640 nm. Because the operation wavelength (640 nm) and the resonant peak wavelength (496 nm) are close to each other, a large EO coefficient value is expected at 640 nm owing to the resonance effect, with a consequently high modulation efficiency. In addition, our EO polymer exhibits a high glass transition temperature (T_g) of 164 °C and excellent thermal stability.

We demonstrated the successful working of our EO-polymer modulator, which has a low half-wave voltage-length product ($V_{\pi}L$) of 0.52 V·cm at $\lambda = 640$ nm. The proposed device features a high modulation efficiency as well as a near-infrared modulator. To the best of our knowledge, this experimentally developed modulator-based EO polymer is the lowest $V_{\pi}L$ modulator operating at wavelengths below 1000 nm reported thus far.

2. EO polymer

We synthesized an EO polymer to fabricate a modulator for wavelengths below 1000 nm; its molecular structure is depicted in Fig. 1(a). The mainchain polymer comprises an adamantane-containing methacrylate polymer, while the side-chain contains TCF as the acceptor unit and aminobenzene as the donor unit, with both units directly connected by a single bond. The T_g value of this polymer was 164 °C.

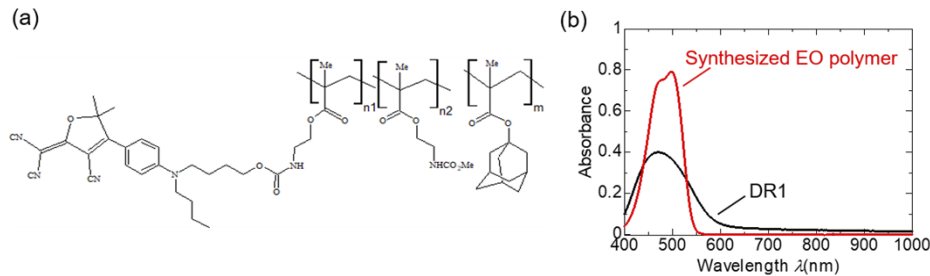


Fig. 1. a Molecular structure of the synthesized EO polymer. b Absorption spectra of the synthesized EO polymer and DR1.

Figure 1(b) depicts the absorption spectrum of the synthesized EO polymer film. The synthesized EO polymer was dissolved in cyclohexanone and spin-coated on a quartz substrate to form a 0.2- μm -thick film. The film was then baked at 170 °C for 1 h under vacuum. As shown in Fig. 1(b), its absorbance was almost zero at wavelengths longer than the red region. The resonance peak wavelength was 496 nm. Figure 1(b) depicts the absorption spectrum of the typical EO polymer, DR1. The absorption of the synthesized EO polymer at 640 nm was sufficiently lower than that of DR1, whose absorption is characterized by a broad band and is discernible at longer wavelengths far from the resonance peak. In contrast, our synthesized EO polymer has a resonance peak tail that drops precipitously at long wavelengths. Additionally, its EO coefficient is high near the resonance peak wavelength; it had a value of 26.8 pm/V at $\lambda = 640$ nm, which was obtained by applying 140 V/ μm at 164 °C for 1 min. These parameters have been chosen to provide sufficient EO coefficients for the modulator. Consequently, the modulator can operate at a wavelength with a high EO coefficient. The optical absorption loss of the EO polymer based on the same chromophore was 2.6 dB/cm at the driven wavelength ($\lambda = 640$ nm) of the modulator. This optical loss was evaluated using thick films of the EO polymer (See Supplement 1, “Evaluation of absorption loss of EO polymer”). The optical absorption loss of the DR1-based EO polymer measured using the same method was 564 dB/cm at $\lambda = 640$ nm. Thus, our synthesized EO polymer has an extremely low loss at the visible wavelength, and it is substantially practical for application in various optical devices used in the visible region.

3. Device design

Figure 2(a) illustrates a schematic of the EO-polymer waveguide of the proposed modulator. A Mach-Zehnder interferometer (MZI) was used to evaluate its modulation properties. The 0.4-cm-long upper electrodes were placed on each arm of the MZI, while the bottom electrode (made of indium zinc oxide, IZO) was placed on the entire surface of the substrate. The light was input from one end-face of the waveguide and the output light was detected from the other. The optical splitter and combiner were implemented using a 1×2 multimode interference (MMI) coupler. The phase of the light wave was modulated by applying it to the upper/bottom electrode, while the amplitude of the traveling wave was modulated by interference at a part of the combiner.

As shown in Fig. 2(b), a ridged waveguide was used. The ridge was 1.0 μm wide and was fabricated by photolithography; the waveguide cladding was an organic silica composite

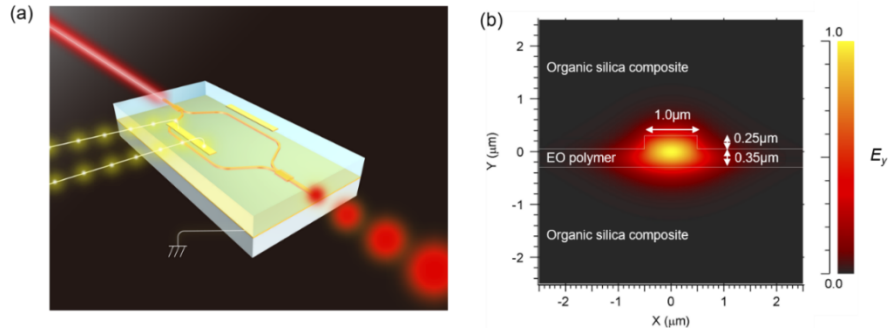


Fig. 2. a Schematic of the intensity-modulator-based EO polymer. Our device is operated at wavelengths in the red region and the intensity of input light is modulated by the electric signal. b Electric mode field of the ridged waveguide.

fabricated using the sol-gel method. Our ridged waveguide was optimized to a single mode ridged waveguide structure (See [Supplement 1](#), “Design of ridged, EO-polymer waveguide”), and Fig. 2(b) depicts its mode field. The confinement light at its core can be evaluated using the confinement factor Γ_c , expressed by

$$\Gamma_c = \int_{core} |\mathbf{E}_y|^2 dy / \int_{\infty} |\mathbf{E}_y|^2 dy \quad (1)$$

Here, E_y denotes the y-axis component of the electric field vector. From Eq. (1), the optimized ridge waveguide was calculated to have a Γ_c of 0.754.

Figure 3(a) depicts the microscopic images of the fabricated MZI modulator (See [Supplement 1](#), “Device fabrication”). A designed MMI was formed at a part of the splitter (Fig. 3(a) inset) and the IZO upper electrodes were placed on the waveguide. A cross-sectional, scanning-electron microscopic (SEM) image of the ridged EO-polymer waveguide is shown in Fig. 3(b), in which its layered structure is clearly seen (IZO as the top and bottom electrodes, the organic silica composite as the top and bottom cladding, and the EO polymer as the waveguide). The width, strip height, and slab thickness of the ridged waveguide were 1.14, 0.27, and 0.33 μm , respectively. The etching process using O_2 and CHF_3 gas was optimized to create a vertical sidewall, which angled at above 85° . The thicknesses of the top and bottom claddings were 1.70 and 1.37 μm , respectively. The propagation loss of the EO-polymer ridged waveguide was 17.3 dB/cm, including the scattering loss of the side wall.

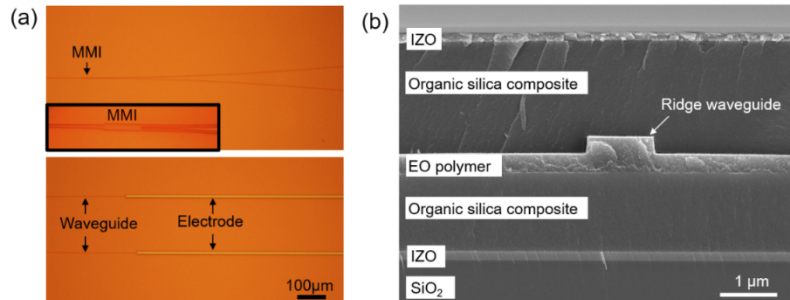


Fig. 3. a Microscopic images of the fabricated MZI using the EO polymer. MMIs (inset) were employed as a beam splitter/combiner. b SEM image of cross-section of phase-shifter-based EO polymer.

4. Modulation characteristics

The modulation properties of the fabricated EO-polymer modulators were evaluated (See [Supplement 1](#), “Modulation measurement”). Figure 4 depicts the optical and drive signals of the EO-polymer modulator. Optical signals typical of the MZ modulator were detected with the driven voltage of the triangular waveform. The maximum and minimum of the optical signal means the relationship in constructive and destructive interference of the light propagating through each arm of the MZI, respectively. The optical signal at applied voltage of 0 V is maximized due to the constructive interference. The peak-to-peak optical signal from the MZ modulator indicated the phase to be π -shifted. The half-wave voltage V_π , directly measured from the applied voltage difference between the minimum and maximum optical signals of the MZI, was determined to be 1.30 V, while the length of phase shifter, L , was 0.4 cm; Together, the voltage-length product, $V_\pi L$, of the modulator was 0.52 V·cm. In general, $V_\pi L$ is a key figure of merit of the optical modulator. The parameters of V_π and L in MZI exhibit a tradeoff. Therefore, the smallest $V_\pi L$ is preferred. The $V_\pi L = 0.52$ V·cm of our device is very small compared to other modulators. Considering absorption loss and efficiency, the operation wavelength of our modulator is approximately from 600 nm to 800 nm.

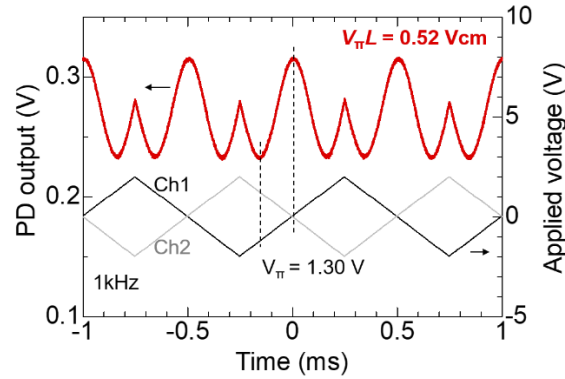


Fig. 4. Optical (red, corrected by a Si photodetector) and drive (black) signals for the measurement of static modulation. Channels 1 (Ch1) and 2 (Ch2) indicate the voltage applied to each arm of the MZI.

5. Discussion

The EO coefficient r_{33} was estimated from the results of the modulation measurements. A multilayer phase modulator was analyzed using the simple equivalent circuit model, where each layer was regarded as a parallel connection of a resistor and capacitor. The modulation driving voltage was applied to each of the three composite layers of the waveguide — the organic silica composite on the top and bottom cladding (represented by subscripts 1 and 3 for thickness and resistivity) and the EO polymer (represented by the subscript 2 for thickness and resistivity) as the core. The thicknesses were set to $d_1 = 1.37 \mu\text{m}$, $d_2 = 0.60 \mu\text{m}$, and $d_3 = 1.70 \mu\text{m}$, while the electrical resistivities were set to $\rho_1 = 555 \times 10^9 \Omega\text{m}$, $\rho_2 = 8.31 \times 10^9 \Omega\text{m}$, and $\rho_3 = 30.2 \times 10^{12} \Omega\text{m}$. The electrical resistivities were evaluated by I-V measurement of individual layers. The permittivity of the EO polymer and the organic silica composite were 2.62 and 3.8, respectively. In this equivalent circuit model, the IZO electrode was ignored because the resistivity of the sputtered IZO film was considerably smaller (approximately $5 \times 10^{-6} \Omega\text{m}$) than that of the EO polymer and organic silica composite. In the calculations of the simple equivalent circuit model, the voltage applied to the EO-polymer layer was estimated to be 0.220 times that applied to

the phase modulator. The value of V_π was 1.30 V, while the electric field applied across the EO-polymer layer was 0.476 V/ μm . The change in the refractive index is expressed as

$$\Delta n = \frac{1}{2} n^3 r_{33} E \quad (2)$$

Additionally, the phase shift at the push-pull operation is expressed as

$$\frac{2\pi}{\lambda} L \Delta n \Gamma = \frac{\pi}{2}. \quad (3)$$

Here, L denotes the length of the phase shifter, and Γ denotes the overlap integral between the applied electric field and optical mode. Because the width of the upper electrode is 10 μm and the applied electric field can be regarded as uniform across the optical mode, the overlap integral Γ is assumed to be nearly equal to the confinement factor Γ_c . In this case, $L = 0.4$ cm and $\Gamma = 0.788$. With these values, Eq. (3) was used to determine r_{33} as 50.8 pm/V.

We compared the performance of our device with that of previously reported modulators, including LN and Si modulators, as depicted in Fig. 5. Our demonstrated modulator is an all-EO-polymer modulator. Previously, all EO-polymer modulators with $V_\pi L = 5.2$ V·cm [45] or 2.4 V·cm [35] in the C-band region in push-pull operation, and $V_\pi L = 4.8$ V·cm [37] or 2.2 V·cm [46] in the O-band region have been reported. However, there are no reports of practical EO-polymer modulators operating at wavelengths below 1000 nm. In LN modulators at communication wavelengths, $V_\pi L = 1.8$ V·cm [29] and 2.5 V·cm [28] at $\lambda = 1550$ nm, and $V_\pi L = 8.2$ V·cm [47] at $\lambda = 1300$ nm were reported. $V_\pi L = 1.6$ V·cm [27] at $\lambda = 850$ nm was reported in LN modulators at wavelengths below 1000 nm. $V_\pi L = 3.2$ V·cm [31] in the C-band region and $V_\pi L = 600$ V·cm [48] at $\lambda = 899$ nm were reported in SiN modulators. Compared to these modulators, our EO-polymer modulator with $V_\pi L = 0.52$ V·cm is extremely efficient. Recently, Si organic hybrid (SOH) modulators including the EO polymer and Si have demonstrated high performances, with $V_\pi L$ values of 0.91 [49] and 0.032 V·cm [33] in the C-band region. Furthermore, a plasmon organic hybrid (POH) using a metal-insulator-metal slot resulted in $V_\pi L = 0.013$ V·cm [50] at $\lambda = 1550$ nm. These SOH and POH modulators use inorganic materials to confine the optical field in a narrow area and narrow gaps to increase the electric field, which is expected to reduce the $V_\pi L$ and reduce the footprint to the sub-micrometer range. In our devices, the modulation efficiency of the EO polymer can be improved by using a hybrid with inorganic materials. The EO-polymer modulator used in this study can also improve the modulation efficiency using waveguide materials such as SiN, TiO₂ [51], Ta₂O₅ [52], and Nb₂O₅ [53], which have transparency windows at visible wavelengths.

The EO coefficient r_{33} of the EO polymer in this study was not as high as that of conventional EO polymers. Previous EO polymers were evaluated using the figure of merit of $n^3 r_{33}$, where n denotes the refractive index and r_{33} represents the EO coefficient. The $n^3 r_{33}$ value of our device was 215.8 pm/V, which is considerably smaller than those reported in previous works, which include values such as 2,300 pm/V [33]. However, the $V_\pi L$ of the proposed modulator is significantly small, which can be attributed to the shorter operating wavelength than those in previous reports. Here, $V_\pi L$ is given by

$$V_\pi L = \frac{\lambda}{n^3 r_{33}} \frac{d}{\Gamma} \quad (4)$$

where λ denotes the wavelength, d denotes the electrode distance, and Γ represents the overlap integral. Modulators with small $V_\pi L$ values are efficient and capable of high-speed operation. Hence, a larger figure of merit, $n^3 r_{33}$, is preferable. The significantly small $V_\pi L$ value despite the small $n^3 r_{33}$ value of this device is due to the small value of d , which indicates the distance between both the electrodes for applying electric voltage to the EO materials. The value of d

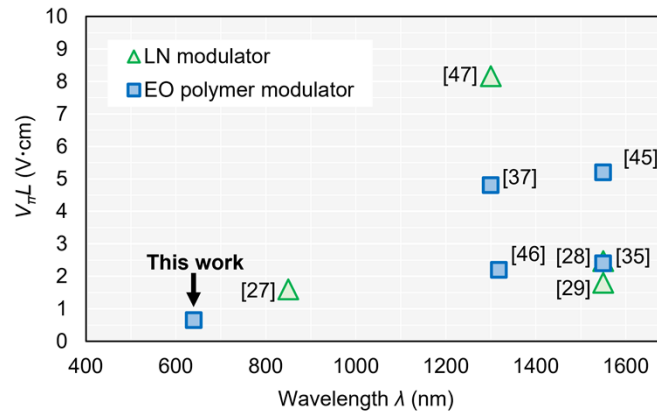


Fig. 5. Comparison of the $V_{\pi}L$ of LN and EO-polymer modulators from visible to NIR wavelengths.

should be designed to suppress the absorption loss of the propagating light. The propagation mode size decreases with a shorter wavelength, and d can be reduced with wavelength. For instance, the mode field diameter (MDF) at $\lambda = 1550$ nm is $2.68 \mu\text{m}$ as shown in Fig. 6(a). Au electrodes were placed above and below the waveguide at a distance d , and the distance d between the electrodes with a loss of 3 dB/cm was calculated using the beam propagation method. As a result, the distance between electrodes d was $4.79 \mu\text{m}$. In contrast, the MDF at $\lambda = 640$ nm was $1.10 \mu\text{m}$ as shown in Fig. 6(b). The distance between the electrodes with a loss of 3 dB/cm was $2.20 \mu\text{m}$ (See Supplement 1, “Design of distance between the electrodes”). In our device, because we synthesized an EO polymer with a significantly high EO coefficient at visible wavelengths, we could demonstrate a modulator with substantially higher efficiency than that of conventional modulators operating at wavelengths above 1000 nm.

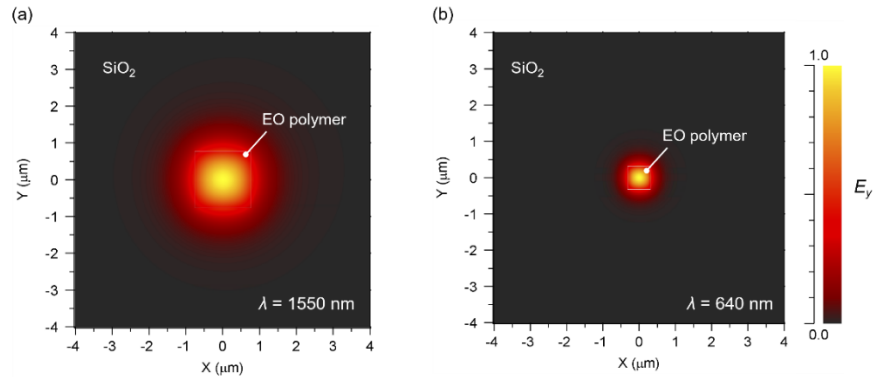


Fig. 6. Mode field of EO-polymer waveguide at a $\lambda = 1550$ nm and b $\lambda = 640$ nm. Waveguide size was optimized for a single mode. The mode field diameters (MDFs) in the y direction are $2.68 \mu\text{m}$ and $1.10 \mu\text{m}$, respectively.

Using an EO-polymer modulator, the fabrication tolerance can be relaxed compared to the slot-type modulator [33], which has a nanoscale structure. Our multilayer structure is easy to fabricate without fabrication errors. Therefore, our designed modulator can also be expanded to form phased arrays that require multiple channel modulators.

The thermal stability of the EO polymer basically depends on the glass transition temperature T_g . The thermal stability of previously reported EO polymers which has similar T_g with our EO polymer had been evaluated [54,55]. In these reports, it has been shown to be sufficiently stable at 80 °C for more than 2000h. Therefore, the same stability can be expected for our EO polymer.

6. Conclusion

In conclusion, a significantly small $V_\pi L$ of 0.52 V·cm was obtained using the EO-polymer modulator. Moreover, this is the first reported successful demonstration of a modulator based on an EO polymer for visible wavelengths. Previous EO-polymer modules have been demonstrated at wavelengths above 1000 nm. We demonstrated that the on-chip modulator operated at $\lambda = 640$ nm, which is a significantly shorter wavelength than the near-infrared wavelength. Thus, the result implies that the device can also operate at wavelengths below the 1000 nm near-infrared region. The wavelengths below 1000 nm of 905 nm and 785 nm are the operating wavelengths of the LiDAR for autonomous driving [23] and excitation light of bio-sensing for the Raman spectroscopy [13,14,56], respectively. In particular, LiDAR consisting of several of our developed phase modulators is expected to be realized soon. Furthermore, visible wavelength photonic platforms are required for neural probes [9,57,58] in optogenetics. Our EO-polymer modulator has the potential for use in future optical on-chip devices. Thus, EO-polymer modulators are expected to become a leading candidate for active photonic devices in visible photonic platforms.

Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

Supplemental document. See [Supplement 1](#) for supporting content.

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